respectively), this difference was not statistically significant (HR 1.02, 95% CI 0.20-5.12, p=0.98).

Primary Histology	Median OS	95% Confidence Interval
All Metastases (n=104)	14.8 months	2.0-27.6
Colorectal mets (n=50)	8.2 months	4.3-12.0
All neuroendocrine mets (n=36)	29.2 months	17.0-41.3
Non-neuroendocrine mets (n=68)	8.7 months	5.7-11.6
Pancreatic neuroendocrine mets (n=12)	41.9 months	15.8-68.1
Non-pancreatic neuroendocrine mets (n=24)	24.6 months	10.9-38.4

Conclusions: Radioembolization with Yttrium-90 microspheres has been shown in previous studies to be a safe and effective treatment for metastases to the liver with some demonstrating a survival benefit over other treatment modalities. Our data confirm the utility of this treatment method in multiple histologies, particularly neuroendocrine primary lesions. Additional prospective dosimetric studies are needed to determine how to optimally escalate Yttrium-90 microspheres in an attempt to further improve survival of these patients.

OR17 Presentation Time: 2:36 PM

Incidence and Treatment of Toxicity following ^{125}I Episcleral Plaque Brachytherapy for Uveal Melanoma

Shahed N. Badiyan, MD[1], Rajesh C. Rao, MD[2], Anthony J. Apicelli, MD, PhD[1], Sahaja Acharya, MD[1], Vivek Verma, BS[1], Adam A. Garsa, MD[1], Todd DeWees, PhD[1], Christina K. Speirs, MD, PhD[1], Jose Garcia-Ramirez, MS[1], Perry W. Grigsby, MD, MS[1], James W. Harbour, MD[3]. [1]*Radiation Oncology, Siteman Cancer Center, Washington University Medical Center, Saint Louis, MO;* [2]*Opthalmology and Visual Sciences, Washington University Medical Center, Saint Louis, MO;* [3]*Ophthalmology, Bascom Palmer Eye Institute, University of Miami Miller School of Medicine, Miami, FL.*

Purpose: To determine the incidence and treatment of ocular toxicity following ^{125}I episcleral plaque brachytherapy.
Materials and Methods: Medical records were reviewed for 469 patients treated with COMS style ^{125}I plaque brachytherapy for uveal melanoma from 1996 to 2011. Ophthalmologic followup records were analyzed to assess the incidence of ocular toxicities after therapy. Rates of toxicities were determined using the Kaplan-Meier method. Fischer's exact test was performed to determine association between date of plaque brachytherapy and need for enucleation due to toxicity.
Results: Mean tumor thickness and long basal diameter were 5.0 mm and 12.1 mm, respectively. The median plaque size was 18 mm. One hundred fourteen plaques (24.3%) required a notch due to proximity to the optic nerve. The median followup was 47 months (range 6-175 months). The median visual acuity in the effected eye prior to treatment was 20/40 and on most recent followup was 20/150. The 3-year rates of radiation retinopathy, radiation papillopathy, and exudative retinal detachment were 45% (n=162), 14% (n=47), and 10% (n=41), respectively. The 3-year rates of cystoid macular edema, vitreous hemorrhage, and enucleation due to radiation toxicity were 17% (n=61), 12% (n=47), and 4% (n=14), respectively. The 3-year rates of cataract formation, neovascularization of the iris and neovascular glaucoma were 29% (n=115), 6% (n=24), and 4% (n=16), respectively. Twenty-nine patients (6.2%) received intravitreal triamcinolone acetonide for treatment of cystoid macular edema, and thirty two patients (6.8%) received subtenons triamcinolone acetonide injections for treatment of cystoid macular edema or radiation retinopathy. Beginning in 2005 patients began receiving intravitreal bevacizumab for treatment of cystoid macular edema, neovascularization of the iris, or neovascular glaucoma. Seventy-one patients (15.1%) received intravitreal bevacizumab. Patients undergoing plaque brachytherapy after 2005 were significantly less likely to require an enucleation due to radiation toxicity than patients undergoing plaque brachytherapy prior to 2005 (p = 0.01).
Conclusions: Radiation retinopathy and cataract formation are common toxicities three years following ^{125}I plaque brachytherapy for uveal melanoma. Patients treated after 2005 were significantly less likely to require an enucleation due to toxicity than patients treated prior to 2005, possibly due to the use of bevacizumab.

OR18 Presentation Time: 2:45 PM

HDR-IORT for Locally Advanced or Recurrent Rectal Cancers: Technique and the Peter Mac Experience

Sarat Chander, MD, FRANZCR[1], Jennifer Tan, FRANZCR[1], Alexander Heriot, MD, MBA, FRCS, FRACS[2], John Mackay, FRACS, FRCS[2], Craig Lynch, MBBS, MMedSci, FRAC[2], Sylvia van Dyk, DAppSc[1], Mathias AB. Bressel, MSc[3], Trevor Leong, MD, FRANZCR[1], Sam Ngan, MBBS, FRCSEd, FRANZCR[1]. [1]*Radiation Oncology;* [2]*Surgical Oncology;* [3]*Biostatistics and Clinical Trials, Peter MacCallum Cancer Centre, Melbourne, VIC, Australia.*

Purpose: Our technique and experience in delivering high-dose-rate intraoperative radiotherapy (HDR-IORT) for locally advanced primary or recurrent rectal cancer using a custom-made brachytherapy applicator is described.
Materials and Methods: Patients selected are those with T4 rectal cancer or pelvic recurrence, deemed suitable for radical surgery but at high-risk of positive resection margins, without evidence of metastasis. Chemo-radiation is followed by radical surgery. The tumor is excised and the boundaries of the tumor bed are marked with surgical clips. A customized applicator is fashioned using layers of self-adhesive skin dressing (Stomahesive®) and flexible catheters, which can be molded to the shape of the tumor bed. Ten gray (Gy) is delivered to tumor bed via a customized IORT applicator at time of surgery.
Results: Twenty-seven patients were treated between Feb 2004 and June 2007. There were 15% primary and 85% recurrent cancers. Seventy-one percent received preoperative chemoradiation to 45-50.4 Gy. R0, R1 and R2 resection were 70%, 22% and 7%, respectively. Irradiated sites included tumor beds (18), nodal regions (2) and wider pelvic areas (12). Ten patients [37% (95% CI=19-58)] experienced grade 3 or 4 toxicities (3 wound, 4 abscesses, 3 soft tissue, 3 bowel obstructions, 3 ureteric obstructions, 2 sensory neuropathies). Local recurrence-free, failure free and overall survival rates at 2.5 years were 68% (95% CI=52-89), 37% (95% CI=23-61) and 82% (95% CI=68-98), respectively.
Conclusions: The addition of IORT, via a customized applicator at the time of surgery appears to be safe and may be beneficial for local control.

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Accurate Calculation of Biological Equivalent Dose from Six Dosimetric Quality Indicators in a Randomized Trial for Intermediate-Risk Prostate Patients

Wayne M. Butler, PhD, Gregory S. Merrick, MD. Schiffler Cancer Center, Wheeling Hospital, Wheeling, WV.

Purpose: To calculate the biologically equivalent dose (BED) using 6 brachytherapy dosimetric quality indicators and compare BEDs between two arms of a randomized trial.
Materials and Methods: All 246 intermediate-risk patients randomized to either a conventional dose of 44 Gy external beam radiotherapy (EBRT) or a reduced EBRT dose of 20 Gy followed by a scaled brachytherapy boost had complete data on 6 dosimetric quality indicators. The trial, closed in June 2004, added 90 Gy ^{103}Pd to 44 Gy EBRT versus 115 Gy ^{103}Pd added to 20 Gy EBRT. Six dosimetric quality indicators, D_{100}, D_{90}, V_{90}, V_{100}, V_{150}, and V_{200}, help define the shape of the brachytherapy dose volume histogram (DVH). Using the radiobiological methodology of AAPM